

Simulated Microplastic Release from Cutting Boards and Evaluation of Intestinal Inflammation and Gut Microbiota in Mice

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BACKGROUND: Plastic cutting boards are commonly used in food preparation, increasing human exposure to microplastics (MPs). However, the health implications are still not well understood.

OBJECTIVES: The objective of this study was to assess the impacts of long-term exposure to MPs released from cutting boards on intestinal inflammation and gut microbiota.

METHODS: MPs were incorporated into mouse diets by cutting the food on polypropylene (PP), polyethylene (PE), and willow wooden (WB) cutting boards, and the diets were fed to mice over periods of 4 and 12 wk. Serum levels of C-reactive protein (CRP), tumor necrosis factor- α (TNF- α), interleukin-10 (IL-10), lipopolysaccharide (LPS, an endotoxin), and carcinoembryonic antigen (CEA), along with ileum and colon levels of interleukin-1 β (IL-1 β), TNF- α , malondialdehyde (MDA), superoxide dismutase (SOD), secretory immunoglobulin A (sIgA), and myosin light chain kinase (MLCK), were measured using mouse enzyme-linked immunosorbent assay (ELISA) kits. The mRNA expression of mucin 2 and intestinal tight junction proteins in mouse ileum and colon tissues was quantified using real-time quantitative reverse transcription polymerase chain reaction. Fecal microbiota, fecal metabolomics, and liver metabolomics were characterized.

RESULTS: PP and PE cutting boards released MPs, with concentrations reaching $1,088 \pm 95.0$ and $1,211 \pm 322$ $\mu\text{g/g}$ in diets, respectively, and displaying mean particle sizes of 10.4 ± 0.96 vs. 27.4 ± 1.45 μm . Mice fed diets prepared on PP cutting boards for 12 wk exhibited significantly higher serum levels of LPS, CRP, TNF- α , IL-10, and CEA, as well as higher levels of IL-1 β , TNF- α , MDA, SOD, and MLCK in the ileum and colon compared with mice fed diets prepared on WB cutting boards. These mice also showed lower relative expression of *Occludin* and *Zonula occludens-1* in the ileum and colon. In contrast, mice exposed to diets prepared on PE cutting boards for 12 wk did not show evident inflammation; however, there was a significant decrease in the relative abundance of Firmicutes and an increase in Desulfobacterota compared with those fed diets prepared on WB cutting boards, and exposure to diets prepared on PE cutting boards over 12 wk also altered mouse fecal and liver metabolites compared with those fed diets prepared on WB cutting boards.

DISCUSSION: The findings suggest that MPs from PP cutting boards impair intestinal barrier function and induce inflammation, whereas those from PE cutting boards affect the gut microbiota, gut metabolism, and liver metabolism in the mouse model. These findings offer crucial insights into the safe use of plastic cutting boards. <https://doi.org/10.1289/EHP15472>

Introduction

Despite recycling initiatives aimed at banning single-use plastics, their pervasive use and inadequate management worldwide have led to the presence of microplastics (MPs) in various environmental compartments, including oceans,¹ freshwater systems,² agricultural lands,³ urban areas,⁴ the atmosphere,⁵ and even remote locations such as Mount Everest.⁶ Humans are unavoidably exposed to MPs, which have been found in human feces,⁷ blood,⁸ lungs,⁹ placentas,¹⁰ and livers.¹¹ Dietary intake is a key contributor to MPs exposure, with MPs inadvertently present in consumed items, such as bottled water,¹² salt,¹³ beer,¹⁴ and canned fish.¹⁵ Based on the number of MPs in foods, studies estimate that Americans ingest >50,000 MP particles annually.¹⁶ Further research suggests individuals might ingest up to 1,530 and 587 MPs daily through food and water, respectively.¹⁷ In addition, plastic utensils, such as polypropylene (PP) infant

feeding bottles¹⁸ and plastic tea bags,¹⁹ can release substantial quantities of MPs, further contributing to human intake.

Researchers are concerned about the health effects of MPs based on factors such as particle size, polymer, shape, charge, concentration, and exposure routes.²⁰ Regulatory bodies, including the European Food Safety Authority²¹ and the World Health Organization,²² have indicated that MPs may pose minimal adverse health effects, although this assessment may be based on limited data rather than an absence of effects. Signs of impact continue to emerge from toxicity studies using rodents. A key proposed mechanism for MP toxicity is the induction of local oxidative stress and inflammation.^{23,24} For instance, mice exposed to MPs in drinking water have exhibited compromised antioxidant defenses, evidenced by reduced superoxide dismutase (SOD) and glutathione (GSH) levels and increased malondialdehyde (MDA) formation.²⁰ Polystyrene (PS) MPs have accumulated in the kidneys and livers of mice, enhancing the activity of the NF-E2-related factor 2/Kelch-like ECH-associated protein 1 (Nrf2/Keap1) pathway and causing inflammation through the promotion of excessive reactive oxygen species.^{25–27} Furthermore, MPs have been shown to disrupt the gut microbiota of mice and humans,²⁸ impairing the balance of intestinal oxidation and inflammation²⁹ and compromising intestinal epithelial barrier functions.³⁰ Increased microbial load and diversity have been observed in fecal samples from mice fed polyethylene (PE) particles (600 $\mu\text{g/d}$ for 35 d).³¹ In addition, altered hepatic bile acid levels and changes in serum bile- and amino acid-related metabolites were noted in mice exposed to 5- μm PS MPs (100 and 1,000 $\mu\text{g/L}$) in drinking water for 6 wk.³² A recent study associating MPs with cardiovascular diseases found that patients undergoing carotid endarterectomy for asymptomatic carotid artery disease, who had MPs and nanoplastics detected in their carotid artery plaque, faced a higher risk of myocardial infarction, stroke,

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or death from any cause at 34 months of follow-up compared with those without detected micro- and nanoplastics (MNPs).³³ This finding highlights inflammation as the likely mechanism linking MPs to health issues.³³ Despite advances in understanding MP toxicity, most studies have used commercial MP pellets, which differ from environmentally derived MPs. To bridge this knowledge gap, further toxicity assessments of environmental MPs are essential.

A cutting board is a fundamental tool in the kitchen. Concerns regarding bacterial contamination in wooden cutting boards have led to the increased adoption of plastic alternatives,^{34,35} increasing the potential for MP transfer and human exposure during food preparation. A single cut on a new PP cutting board may release 100–300 MPs.³⁶ Research has estimated annual per-person exposure to be 7.4–50.7 g of MPs from PE cutting boards and 49.5 g of MPs from PP cutting boards.³⁷ Although studies have directly measured MPs in foods originating from plastic cutting boards,^{38,39} no research has thoroughly examined the health effects of exposure to these MPs.

In this study, mouse diets (AIN 93G) were prepared by cutting on PP, PE, and willow wooden (WB) cutting boards to produce 12 batches of feed for each type of cutting board. The MPs in each batch of diets were both qualitatively and quantitatively characterized before being administered to mice for periods of 4 and 12 wk, during which intestinal inflammation, fecal microbiota and metabolomics alteration, and liver metabolomics alteration were assessed. The objective of this research was to evaluate the effects of long-term exposure to MPs released from various cutting boards on intestinal inflammation and gut microbiota using mouse bioassays. This study addresses an important issue concerning the safety of plastic cutting boards and their role in MP exposure. The findings provide significant insights into how MPs from household items impact health, particularly regarding intestinal inflammation and gut microbiota.

Materials and Methods

Cutting Boards and Characterization

PP and PE cutting boards have been previously assessed for MP release.³⁷ To investigate the release of MPs and their associated health effects, new cutting boards made of PP and PE (brands: Yongyou and Kameilai, dimensions: 36 × 25 cm and 40 × 30 cm, respectively; $n = 2$ for each type) were acquired. For control purposes, new WB cutting boards (brand: Yaotai, dimensions: 45 × 32 cm, $n = 2$) were also purchased.

To characterize potential heavy metal contaminants and additives in cutting boards, the lead researcher (H.-J.G.) conducted 7,000 cuts on these boards using a stainless steel kitchen knife (brand: Fujun, blade length: 15.6 cm, handle height: 9.5 cm) without food. The cutting force of each strike, ranging between 200 and 600 N, was monitored in real time using a three-dimensional (3D) force measurement platform (Bertec FP4060-PT) at One Measurement Group Limited in Suzhou, China. Following cutting, the released particles were scraped from the cutting boards, weighed to 0.1 g, and underwent triplicate digestion with 10 mL of nitric acid (HNO₃) and 2 mL of hydrogen peroxide (H₂O₂) at 250°C for 4 h. The digested solutions were diluted with Milli-Q water, filtered through a 0.22-μm polyethersulfone (PES) syringe filter (GREEN MALL, Jiangsu Green Union Science Instrument Co., Ltd.), and subsequently analyzed for antimony (Sb), arsenic (As), cadmium (Cd), cobalt (Co), chromium (Cr), lead (Pb), and nickel (Ni) using inductively coupled plasma mass spectrometry (ICP-MS; NexION300, PerkinElmer).⁴⁰ In addition, 0.1 g of particles were extracted twice with 5 mL of acetone each time through sonication for ~30 min.⁴¹ After centrifugation at

2,500 rpm for 20 min, the supernatants were collected, concentrated to ~0.5 mL under nitrogen and adjusted to 1 mL gravimetrically with *n*-hexane. These extracts were then filtered through 0.22-μm PES syringe filters and analyzed for seven phthalates: dibutyl phthalate (DBP), diisobutyl phthalate (DIBP), dicyclohexyl phthalate (DCHP), bis(2-ethylhexyl) phthalate (DEHP), di-*n*-octyl phthalate (DNOP), butyl benzyl phthalate (BBP), and dihexyl phthalate (DNHP) using gas chromatography–MS (GC-MS; 7890A, Agilent Technologies). The initial column temperature was 60°C, which was held for 5 min and was subsequently raised to 300°C at a rate of 15°C/min, and held for 8 min. The injector temperature was set at 280°C. Helium was used as the carrier gas at a flow rate of 1 mL/min.

Incorporation of MPs into Mouse Diets through Cutting

To evaluate, in mice, the health effects of MP exposure from the daily use of plastic cutting boards, 12 batches of diet pastes—each consisting of 160 g of pulverized AIN 93G diet powder (Jiangsu Medicine Ltd.) and 20 mL of Milli-Q water—were successively cut on PP, PE, or WB cutting boards. The authors performed these cuts using a stainless steel kitchen knife, executing 700 strikes per batch with an average cutting force of 300 N, which incorporated released plastic or wooden particles into the diets (Figure 1). The modified diets were then freeze-dried and provided to the mice for consumption over periods of 4 and 12 wk (Figure 1).

Initially, during the first cutting cycle, the AIN 93G diet was shaped into a pancake ~16 cm in diameter and manually cut on an intact cutting board using 700 knife strikes, evenly divided between one direction and a perpendicular orientation (Figure 1). This number of cuts was selected to mimic the typical frequency of cutting in a kitchen over 1 wk. For each board, the cutting force of every strike was monitored in real time by a 3D force-measuring platform (Bertec FP4060-PT) in One Measurement Group Limited. A video demonstrating the cutting process is available in the Supplemental Material (“Cutting video”). Despite variations in the force of each strike, ranging from 200 to 600 N, no significant difference was observed in the overall force applied across the three types of cutting boards, which averaged 300 N (Figure S1). This force was applied to simulate the typical cutting force used in preparing minced meat. For each type of cutting board, duplicate diets were prepared on two cutting boards (i.e., 2 × 160 g), and the resulting feeds from both cutting boards were thoroughly mixed, divided into subsections, rolled into balls, freeze-dried, and then provided to 12 mice over the first week. Preparing 320 g of diet ensured a daily intake of ~3.5 g per mouse for 12 mice over 7 d.

After the initial cutting cycle, the cutting boards were thoroughly cleaned with ultrapure water, air-dried, and then subjected to the second cycle of feed preparation. This involved cutting a new batch of AIN93G paste in a manner similar to the first cycle, thereby producing diets for mouse consumption during the second week. The process of cutting and preparing diets on the cutting boards was repeated across 12 cycles, yielding 12 batches of feed for continuous mouse exposure over 12 wk. Throughout the entire process, the lead researcher (H.-J.G.) performed the cutting with consistent force.

Characterizing MPs in Mouse Diets

To detect MPs in prepared mouse diets, the diets were digested using Fenton’s reagent and HNO₃, following the protocol of Yan et al.⁷ Briefly, ~5 g of dry weight feed samples were placed in 1-L glass beakers, covered with glass dishes to prevent contamination. Then, 70 mL of Fenton’s reagent [30% H₂O₂ and an iron catalyst solution (20 g of iron(II) sulfate heptahydrate in 1 L of

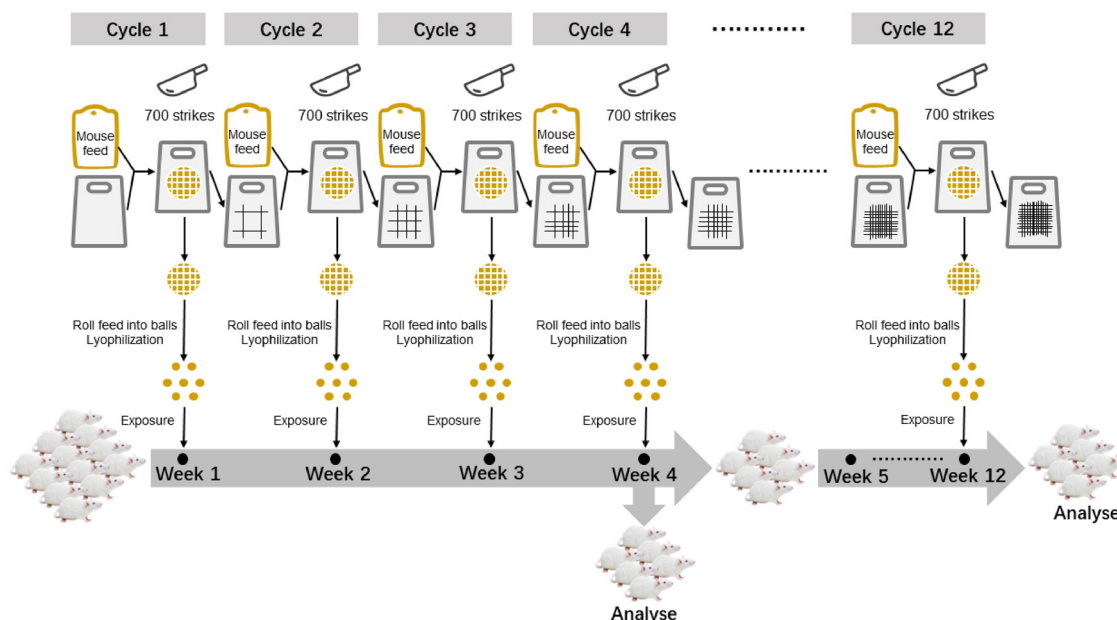


Figure 1. Flow chart showing the preparation of AIN-93G diet on cutting boards (polypropylene, polyethylene, willow wooden) to prepare diets for mouse exposure studies for 12 wk. In the first cutting cycle, 160 g of AIN 93G diet was added to 20 mL of ultrapure water to make a paste, which was spread and cut on an intact cutting board for 700 strikes. For each type of board, diets were prepared in duplicate (i.e., 2×160 g on two cutting boards) with the resulting diets thoroughly mixed, divided into subsections, rolled into balls, freeze-dried, and supplied to 12 mice over the first week, and the cutting board was subject to the next cycle of feed preparation for the second week. Cutting and preparation of mouse diets were repeated for 12 cycles, which allowed for continuous mouse exposure over 12 wk. At the end of the fourth week, 6 of 12 mice were randomly selected for analysis, whereas exposure for the remaining mice continued until the end of the 12th week.

ultrapure water) at a volume ratio of 2.5:1] was added to digest the feed samples at temperatures $<40^{\circ}\text{C}$ for 30 min. This digestion process was repeated twice more by adding an additional 70 mL of Fenton's reagent each time. The total digestion time was kept under 5 h to prevent the destruction of the plastic particles. The solutions were subsequently filtered using mixed cellulose nitrate and cellulose acetate (CN-CA) filters (47-mm diameter, $1\text{-}\mu\text{m}$ pore size). The CN-CA filters were then placed in 1-L glass beakers and further digested with 50 mL of 65% HNO_3 in a 65°C water bath for 1 h to digest the remaining diet components. The resulting solution was diluted with ultrapure water at a volume ratio of 1:2 and filtered through preweighed poly(tetrafluoroethylene) (PTFE) filters (47-mm diameter, $1\text{-}\mu\text{m}$ pore size). The PTFE filters were rinsed with absolute ethyl alcohol, oven-dried at 65°C , and reweighed. The weight difference was considered the weight of MPs collected on the PTFE filters, which was used to calculate the MP concentration in the test feed samples. The 65% HNO_3 , 30% H_2O_2 , and ethyl alcohol of analytical purity were obtained from Sinopharm Chemical Reagent Co., Ltd. The CN-CA filters and PTFE filters were obtained from Jiangsu Green Union Science Instrument Co., Ltd.

During digestion, AIN 93G diets spiked with commercially available PP and PE MPs (30 μm , produced by Shanghai Youngli Electromechanical Technology Co., Ltd., PP-36080 and PE-50200) at 1,000 $\mu\text{g/g}$ were also included, yielding a concentration recovery rate of 84.9%–102% using this method. To determine whether MPs from the experimental environment contaminated samples during handling, we also disintegrated feeds prepared on wooden cutting boards and characterized and quantitated the particles collected on PTFE filters as described above.

Particles collected on PTFE filters were scraped off and then assessed for composition using a Fourier transform infrared spectrometer (FTIR; Thermo Scientific, Nicolet iS50 FT-IR) in attenuated total reflection mode. Spectra were acquired over a range

of 4,000–400/ cm , with 64 scans, and plotted and analyzed using Origin (version 2022; OriginLab Corporation). Prior to analyses using scanning electron microscopy (SEM), the particles collected on the PTFE filters were reeluted into anhydrous ethanol, added dropwise to clean tinfoil, and allowed to dry naturally. The operation was repeated several times so that there were enough particles on the tinfoil, which were then analyzed for surface morphology and particle size using SEM. Specifically, five locations were randomly selected within the electron microscope field of view of each sample. At each location, 10 plastic particles were randomly chosen, and their minimum diameters were measured to characterize the distribution of particle sizes.

MP Exposure to Mice

After the diet preparation, a mouse bioassay was conducted to assess the health impacts of exposure to MPs incorporated into the diets. The experiments received approval from the Nanjing University Committee on Animal Care. Initially, 36 female Balb/c mice, aged 6 wk and weighing 20–23 g, were divided into three groups of 12 and fed diets prepared on PP, PE, or WB cutting boards during the first cutting cycle. Each mouse was housed individually in a cage under standard animal house conditions, including a 12 h light/dark cycle, a temperature of 25°C , and 50% humidity, and allowed free feed consumption over the first week (Figure 1). At the end of the week, the remaining feed was collected, and the diet consumption per animal was estimated by calculating the weight difference between the feed supplied and the feed remaining.

The 12 mice in each group were subsequently fed diets prepared on corresponding cutting boards during the second, third, and fourth cutting cycles over the second, third, and fourth week, respectively (Figure 1). At the end of the fourth week, 6 mice from each group were randomly selected for analysis. The remaining 6 mice in each group continued to be exposed to diets

prepared on corresponding cutting boards from the 5th to the 12th cycle over the subsequent 8 wk. Mice were euthanized with carbon dioxide (CO₂) at the end of the 4th and 12th weeks to assess temporal changes in health biomarkers following exposure to the released MPs.

At each sampling time point, mouse blood was collected into coagulation-promoting tubes following eyeball removal and centrifuged for 10 min at 4,000 rpm to obtain serum samples for inflammation biomarker analyses. The liver from each mouse was collected and flash-frozen in liquid nitrogen for metabolomics analyses. The mouse duodenum, jejunum, ileum, cecum, and colon were dissected. Cecal contents from each mouse were collected and flash-frozen in liquid nitrogen for fecal microbiota and metabolomics analyses. In addition, ~1-cm ~2-cm sections of ileum and colon tissues were quickly collected and flash-frozen in liquid nitrogen for further inflammation biomarker analyses. Furthermore, ~1 cm sections of colon tissue were immediately fixed in ethanol–Canoy fixative for Alcian Blue and Periodic Acid–Schiff (AB-PAS) staining to examine the mucous layer, a protective barrier on the intestinal epithelial surface that separates the gut microbiota from the intestinal epithelium.⁴² All biological samples were collected in six replicates and stored at –80°C, except for the colon tissues fixed in ethanol–Canoy fixative.

Characterizing MPs in Mouse Tissues

To detect MPs in the mouse tissues, liver, jejunum, and colon samples of mice fed the diets prepared on plastic cutting boards over 12 wk were lyophilized to a constant weight. These mouse tissue samples were then digested by adding 67% nitric acid in a 1:10 vol ratio for 48 h at room temperature, followed by further digestion at 110°C for ~3 h. The solution was evaporated and concentrated to 1 mL at 110°C. Next, 70 µL of the solution was concentrated 3-fold into a pyrolysis cup and completely dried at 110°C. The dried samples prepared from the mouse tissues were then analyzed for pyrolyzed products using pyrolysis-GC-MS (Py-GC-MS; Trace ISQ). To serve as detection references, plastic particles scraped from the PP and PE cutting boards were also subjected to analyses. The samples underwent pyrolysis at 600°C in a Frontier Lab EGA/PY-3030D instrument, and the pyrolyzed products were directly injected into a Trace ISQ instrument equipped with a TR-5MS column (30 m × 0.25 mm × 0.25 µm, Restek) for separation. The column temperature was programmed to increase from 40°C (held for 2 min) at a rate of 20°C/min to 320°C (held for 2 min), totaling an 18-min program. Helium was used as the carrier gas at a flow rate of 1 mL/min. The ion source temperature was set at 230°C, and the mass range analyzed was 29–600 *m/z*.^{8,43,44} The detection of PP MPs in mouse tissues was confirmed by identifying 2,4-dimethyl-1-heptene, a characteristic PP degradation product, in the pyrolyzed samples' chromatograms. Similarly, the presence of PE MPs was verified by detecting two characteristic degradation products, 1-pentadecene and 1-dodecene, from PE in the chromatograms.

Serum, Ileum, and Colon Biochemical Analyses

To assess inflammation, serum levels of C-reactive protein (CRP), pro-inflammatory cytokine tumor necrosis factor- α (TNF- α), anti-inflammatory cytokine interleukin-10 (IL-10), and the endotoxin lipopolysaccharide (LPS) were measured using enzyme-linked immunosorbent assay (ELISA) kits (MM-0074M1, MM-0132M1, MM-0174M1, and MM-45080M1; Jiangsu Meimian Industrial Co., Ltd.). To assess the potential for cancer, the serum level of carcinoembryonic antigen (CEA, a tumor biomarker) was

determined using an ELISA kit (CB10395-Mu, Shanghai Coibo Bio Technology Co., Ltd.).

To assess intestinal oxidative stress and immune response, 0.2 g of ileum and colon tissues were homogenized in a phosphate-buffered solution at a ratio of 1:9 (g:mL) ratio. The homogenates were centrifuged for 10 min at 4,000 rpm and 4°C, and the supernatants were collected for assessing IL-1 β , TNF- α , and myosin light chain kinase (MLCK) using mouse ELISA kits (MM-0040M1, MM-0132M1, and MM-46939M1; Jiangsu Meimian Industrial Co., Ltd.). Levels of MDA, SOD, and secretory immunoglobulin A (sIgA) were also assessed using mouse ELISA kits (CB10205-Mu, CB10221-Mu, and CB10228-Mu; Shanghai Coibo Bio Technology Co., Ltd.). Protein content was determined using an enhanced bicinchoninic acid (BCA) protein assay kit (A045-4A, Nanjing Jiancheng Bioengineering Institute), which served as a reference of IL-1 β , TNF- α , MLCK, MDA, SOD, and sIgA levels.

Colon Histopathology

To demonstrate the effects of exposure to released MPs on the mucous barrier, mouse colon tissue sections fixed in ethanol–Canoy fixative were processed through a series of steps: dehydration in ethanol, clearing in xylene, embedding in paraffin wax, and sectioning into 5-µm thick slices using an HM 340E Rotary Microtome (Thermo Scientific). These sections were then stained with AB-PAS and imaged using a Panoramic Scan (3DHISTECH). Six colon tissue samples, with five sections per sample stained, were analyzed for each group. To quantify the mucous layer, the mucus coverage ratio was calculated as the ratio of mucous area pixels to the total pixel area of the gut section, using Image Pro Plus software (version 6.0; Media Cybernetics).

RNA Isolation and Quantitative Real-Time Polymerase Chain Reaction

The expression of genes coding for mucin 2 protein (MUC2, a major component of intestinal mucus) and intestinal tight junction proteins [Occludin, Claudin-4, and Zona Occludens 1 (ZO-1)], which are critical for controlling intestinal permeability, was quantified in ileum and colon tissues of mice using real-time quantitative reverse transcription polymerase chain reaction (qRT-PCR). Briefly, total RNA was extracted and purified from fresh ileum and colon sections (~0.2-cm long) using the FastPure Cell/Tissue Total RNA Isolation Kit V2 (Nanjing Vazyme Biotech Co., Ltd.). RNA concentration and purity were measured using a Nanodrop spectrophotometer (NanoDrop One, Thermo Scientific). Reverse transcription of RNA (1 µg) to complementary DNA (cDNA) was conducted using a HiScript III RT SuperMix for qPCR (+gDNA wiper) Kit (Nanjing Vazyme Biotech Co., Ltd.). qRT-PCR was performed using Hieff UNICON qPCR SYBR Green Master Mix (Cat#11198ES08, Yeasen Biotech CO., Ltd.) on a Bio-Rad Real-Time PCR System (T100 Thermal Cycler). PCR amplification cycling conditions were as follows: predenaturation (95°C for 3 min), 40 cycles consisting of denaturation (95°C for 10 s), and primer annealing and extension (60°C for 30 s). All the qRT-PCR primers are listed in Table S1, purchased from Tsingke Biotechnology Co., Ltd. The mRNA expression data are expressed using the 2^{– $\Delta\Delta C_t$} method.

Characterizing Fecal Microbiota

Intestinal microbiota serve as the first barrier in the host's gut.⁴² To assess the effects of exposure to released MPs on fecal microbiota, cecal content samples (six replicates per group) were individually extracted using a Stool Genomic DNA kit (CoWin Biotech). The extracted DNA was amplified using PCR (GeneAmp9700, ABI) targeting the bacterial V3–V4 region of 16S rRNA using primer

pairs 338F (5'-ACTCCTACGGGAGGCAGCAG-3')/806R (5'-GGACTACHVGGGTWTCTAAT-3'). The PCR reaction mixture consisted of 4 μ L 5 $\times$ Fast Pfu buffer, 2 μ L 2.5 mM deoxynucleotide triphosphates (dNTPs), 0.8 μ L each primer (5 μ M), 0.4 μ L Fast Pfu polymerase, 10 ng of template DNA, and double-distilled water to a final volume of 20 μ L. PCR amplification cycling conditions included an initial denaturation at 95°C for 3 min, followed by 27 cycles of denaturing at 95°C for 30 s, annealing at 55°C for 30 s, and extension at 72°C for 45 s, with a final extension at 72°C for 10 min, and a hold at 4°C. At the 5' end of the 806R, a unique 8-nt barcode was added to differentiate samples. The PCR products were purified using the AxyPrep DNA Gel Extraction Kit (Axygen Biosciences) and quantified with a Quantus Fluorometer (Promega). The purified amplicons were pooled in equimolar amounts and sequenced using paired-end sequencing on an Illumina MiSeq PE300 platform (Majorbio Bio-Pharm Technology Co., Ltd.). The raw sequence data were deposited in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (PRJNA1095893).

Using Quantitative Insights into Microbial Ecology (QIIME 2; version 2020.2; <https://qiime2.org>), sequence reads were clustered into operational taxonomic units (OTUs) using Uparse (version 11) with a 97% similarity cutoff, following quality-filtering and merging by Fastp (version 0.19.6, HaploX), Usearch (version 11, https://drive5.com/usearch/manual/uparse_pipeline.html), and FLASH (version 1.2.11, <https://ccb.jhu.edu/software/FLASH/>).⁴⁵ The beta diversity of microbial communities was calculated using both weighted and unweighted UniFrac methods. Principal coordinate analysis (PCA) was conducted by ropls [R packages (version 1.6.2; R Development Core Team)] to explore differences in microbial communities between groups, based on a Bray–Curtis dissimilarity matrix. Differences in the relative abundance of microbes were visualized by heatmaps using scipy in Python (version 1.0.0).

To assess the stability of microbial communities, network analysis based on Spearman correlation was conducted for the OTUs with the top 200 absolute abundances. Only correlations with a coefficient of $r \geq 0.5$ and $p < 0.01$ were retained to construct single-factor correlation networks using Gephi (version 0.9.2).⁴⁶ The stability of microbial networks was evaluated by analyzing network properties, including modularity and the ratio of negative-to-positive associations between taxa. Communities characterized by higher modularity and fewer and more negative associations are considered more stable and less likely to shift in composition under environmental stress. This stability is attributed to positive interactions that can create feedback loops between taxa, which may destabilize microbial communities by affecting the fitness of many taxa dependent on these feedback loops in response to environmental perturbations.⁴⁶

Characterizing Fecal and Liver Metabolites

Fecal and liver metabolomic analyses were conducted on mice fed diets prepared from three different types of cutting boards over 12 wk. Fresh cecal contents and liver samples were analyzed at Shanghai Majorbio Bio-Pharm Technology Co. Ltd., where metabolites were extracted and analyzed using ultra-performance liquid chromatography (LC)–tandem Fourier transform MS (Thermo Q-Exactive) equipped with an electrospray ionization source operating in both positive and negative ion modes. Briefly, 50 mg of fresh cecal contents or liver were weighed into 1 mL of extraction solution (methanol:acetonitrile: water = 2:2:1 with an isotopically labeled internal standard mixture). The samples were then extracted at 35 Hz for 4 min and sonicated in an ice water bath for 5 min. The process was repeated three times. After incubation at –40°C for 1 h, the samples were centrifuged

at 12,000 rpm for 15 min at 4°C. The resulting supernatant was transferred to fresh glass vials and evaluated using LC–tandem MS (LC-MS/MS). Quality control samples were prepared by mixing equal amounts of supernatant from all samples. The mobile phase contained 25 mM ammonium acetate, 25 mM ammonium hydroxide, and 5% acetonitrile. The autosampler temperature was 4°C and the injection volume was 3 μ L. Spectra were collected in negative ion mode at a spray voltage of –3.2 V. The raw data from LC-MS/MS analyses were preprocessed using Progenesis QI software (Waters Corporation). Metabolites were identified based on the Human Metabolome Database (HMDB; version 5.0, <http://www.hmdb.ca/>).⁴⁷ Partial least squares discriminant analysis (PLS-DA) was performed using the R package ropls (version 1.6.2) to differentiate fecal and liver metabolic profiles between groups. Differential metabolites were identified based on variable importance in projection (VIP > 1) from the OPLS-DA model and p -values ($p < 0.05$) from Student's t -tests. The Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway database (release 2017-05-01) was used to assess functional differences associated with these metabolites.

Quality Assurance/Quality Control and Data Processing

All labware was washed three times with ultrapure water to prevent plastic contamination. All reagents were filtered through 0.22- μ m PTFE filters before use. Laboratory attire consisting of plastic-free materials and cotton gloves was worn during all experiments to ensure purity. There was no presence of plastic particles in the diets prepared on WB cutting boards during the whole process of experiments, indicating no plastic contamination. Results are presented as arithmetic means $\pm$ standard deviation. Statistical analyses were conducted using GraphPad Prism (version 9.0.0) software, employing one-way analysis of variance (ANOVA) with Tukey's test ($n = 6$) at $p < 0.05$ for general comparisons. The Wilcoxon test was used to evaluate statistically significant differences ($p < 0.05$) in species abundance within the fecal microbiota between groups.

Results

Heavy Metals and Plastic Additives in Plastic Boards

To characterize potential heavy metal contaminants and additives in cutting boards, the lead researcher conducted 7,000 cuts on these cutting boards using a stainless steel kitchen knife without food (Figure S1), and the scrapings were analyzed using ICP-MS and GC-MS. Low concentrations of Sb, As, Cd, Co, Cr, Pb, Ni, and selenium (0.38–0.42, 0.42–0.73, 0.03–0.11, 0.16–0.24, 2.82–4.11, 5.55–7.55, 4.13–6.71, and 0.53–0.68 μ g/g) were detected in PP and PE cutting boards (Table S2). Concentrations of DBP, DCHP+DEHP, DIBP, and DNOP (0.23–0.49, 0.02–0.11, 0.25–0.39, and 0.003–0.028 μ g/g) were also low in the same boards (Table S3).

MPs in Mouse Diets

Following the digestion of mouse diets prepared on PP and PE cutting boards, particles collected on PTFE filters exhibited FTIR peaks indicative of PP and PE, respectively (Figure 2A,B). SEM analysis showed irregular particle shapes (Figure 2C,E), with released PP particles displaying a smaller mean particle size compared with PE ones (10.4 ± 0.96 vs. 27.4 ± 1.45 μ m) (Figure 2D,F). Diets prepared on WB cutting boards showed no characteristic peaks for PP and PE (Figure S2).

Concentrations of PP and PE MPs in diets were 467 ± 209 and 340 ± 237 μ g/g, respectively, following the first cutting. These concentrations increased in subsequent cycles, reaching $1,088 \pm 95.0$

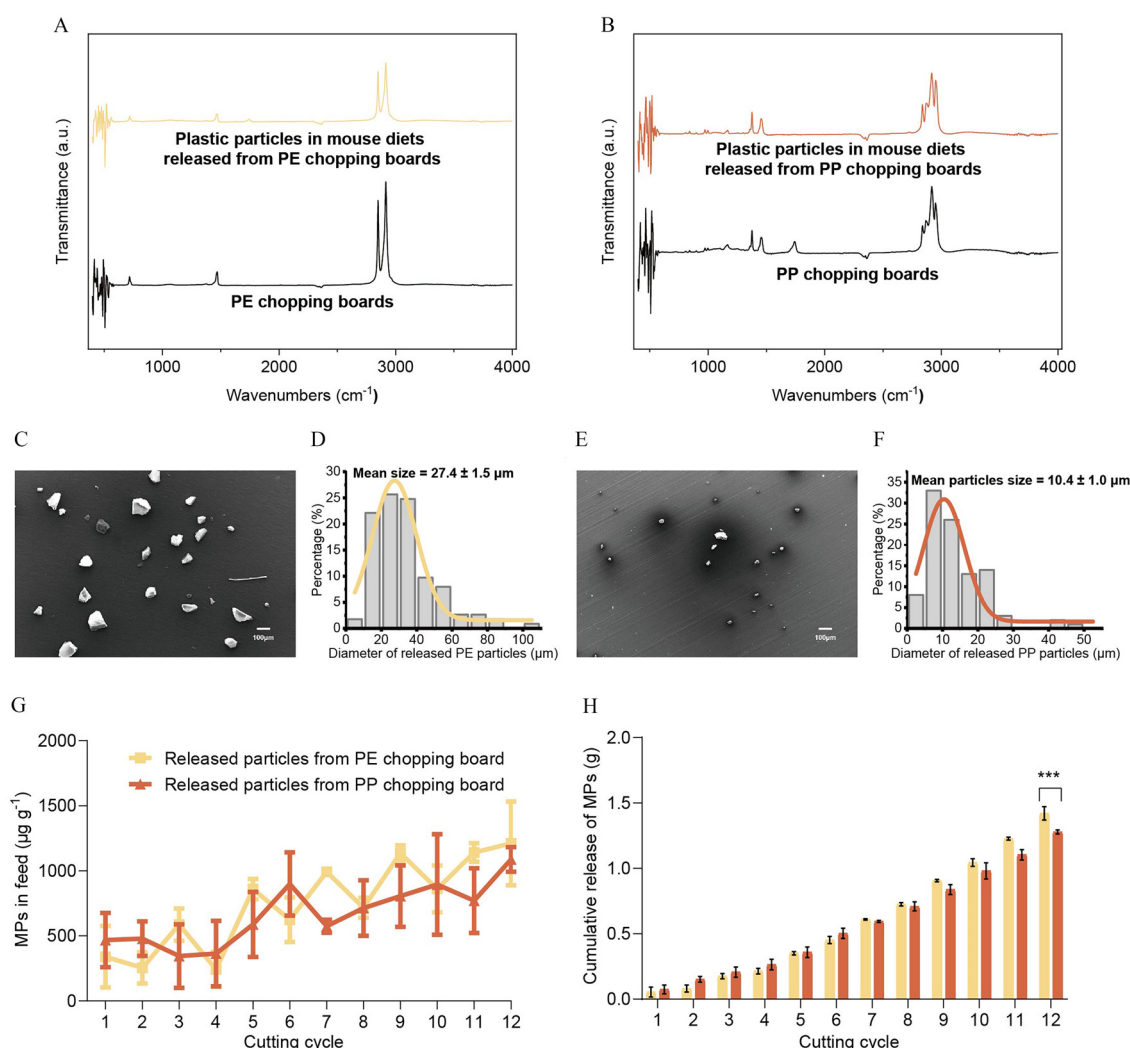


Figure 2. Characterization and quantification of microplastics (MPs) in mouse diets released from polyethylene (PE) and polypropylene (PP) cutting boards. (A) The FTIR spectra for PE cutting boards and plastic particles in mouse diets released from PE cutting boards. (B) The FTIR spectra for PP cutting boards and plastic particles in mouse diets released from PP cutting boards. Scanning electron microscopy images showing shapes of plastic particles released from (C) PE and (E) PP cutting boards. Statistical distribution of particle sizes of plastic particles released from (D) PE and (F) PP cutting boards. (G) Concentration of PE and PP MPs in mouse diets over 12 cycles of diet preparation on PE and PP cutting boards (mean $\pm$ standard deviation, $n = 3$). (H) Cumulative release of PE and PP MPs during 12 cycles of diet preparation on PE and PP cutting boards. Statistical analyses were conducted using GraphPad Prism (version 9.0.0), employing one-way ANOVA with Tukey's test. *** $p < 0.001$. The corresponding data are presented in Excel Table S1. Note: ANOVA, analysis of variance; a.u., absorbance units; FTIR, Fourier transform infrared spectrometry.

and $1,211 \pm 322 \mu\text{g/g}$ by the 12th cycle (Figure 2G). The total accumulated mass of PP and PE MPs over the 12 cutting cycles was 1.28 ± 0.01 and 1.42 ± 0.05 g, respectively (Figure 2H).

Growth of Mice

By supplying diets prepared on WB, PE, and PP cutting boards to mice over 12 wk, no significant difference was observed ($p < 0.05$) in the total feed consumption or body weight gained among the mice of the different groups (Figure S3). These findings indicate MPs released from cutting boards did not measurably alter dietary intake or growth outcomes under the experimental conditions.

MPs in Mouse Tissues

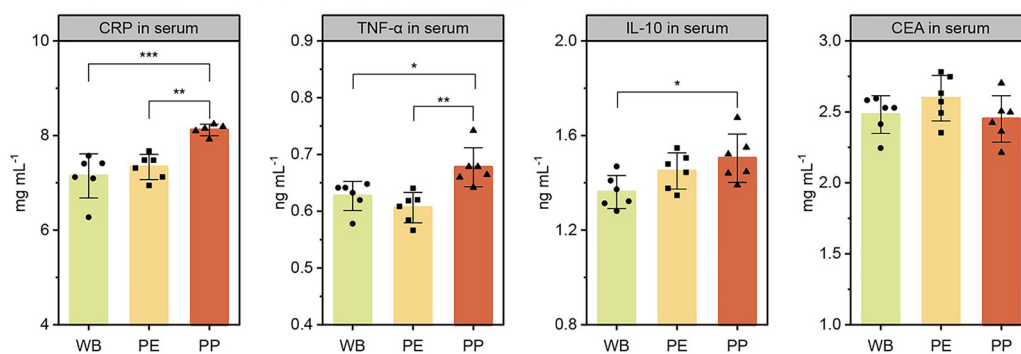
After 12 wk of exposure to diets prepared on PP and PE cutting boards, characteristic degradation products 2,4-dimethyl-1-heptene from PP and 1-pentadecene and 1-dodecene from PE were not detected in the chromatograms of the pyrolyzed mouse liver,

jejunum, and colon samples (Figure S4). The absence of these compounds in the analyzed tissues suggests that exposure to diets prepared on PP and PE cutting boards under the tested conditions did not result in detectable bioaccumulation of the microplastics.

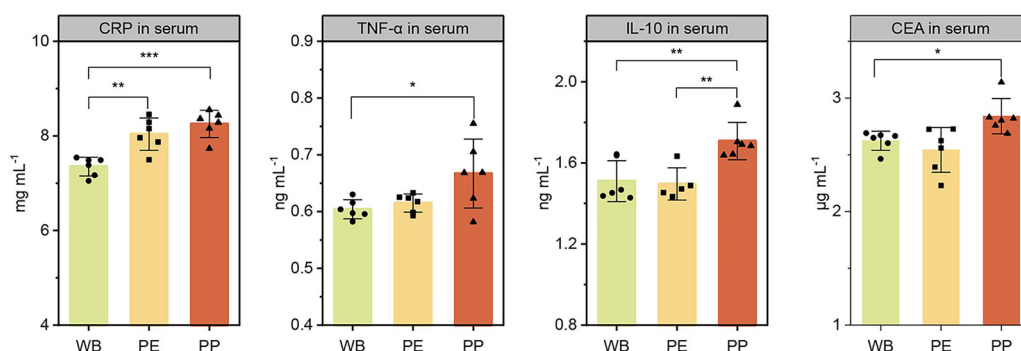
Intestinal Inflammatory Biomarkers

Compared with mice fed diets prepared on WB cutting boards, the mice fed diets prepared on PP cutting boards exhibited significantly ($p < 0.05$) higher serum levels of CRP, TNF- α , and IL-10 after either 4 wk (8.12 ± 0.12 mg/mL, 0.68 ± 0.03 ng/mL, and 1.50 ± 0.10 ng/mL vs. 7.14 ± 0.47 mg/mL, 0.63 ± 0.02 ng/mL, and 1.36 ± 0.07 ng/mL, respectively) or 12 wk (8.25 ± 0.29 mg/mL, 0.67 ± 0.06 ng/mL, and 1.71 ± 0.09 ng/mL vs. 7.35 ± 0.20 mg/mL, 0.60 ± 0.02 ng/mL, and 1.51 ± 0.10 ng/mL, respectively) (Figure 3A,B). In addition, mice fed diets prepared on PP cutting boards over 12 wk exhibited significantly higher serum CEA (2.84 ± 0.16 vs. $2.62 \pm 0.08 \mu\text{g/mL}$, $p = 0.0112$) (Figure 3B); ileum IL-1 β (47.1 ± 12.0 vs. 32.0 ± 7.01 pg/mg protein, $p =$

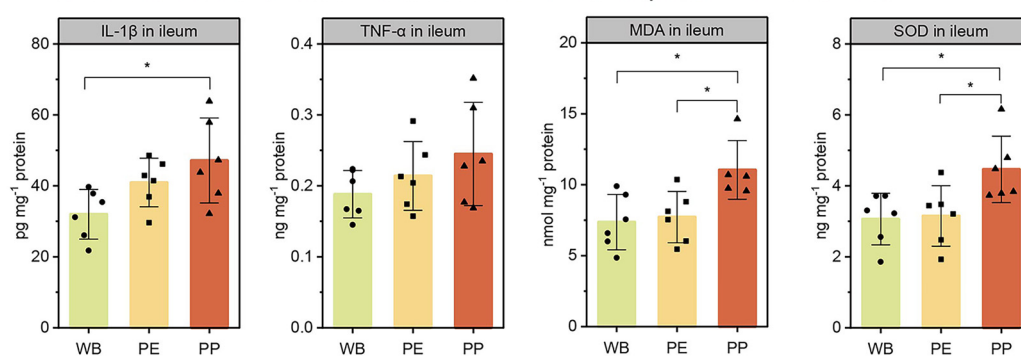
A Inflammation in serum after exposure for 4 weeks



B Inflammation in serum after exposure for 12 weeks



C Inflammation and oxidative stress in ileum after exposure for 12 weeks



D Inflammation and oxidative stress in colon after exposure for 12 weeks

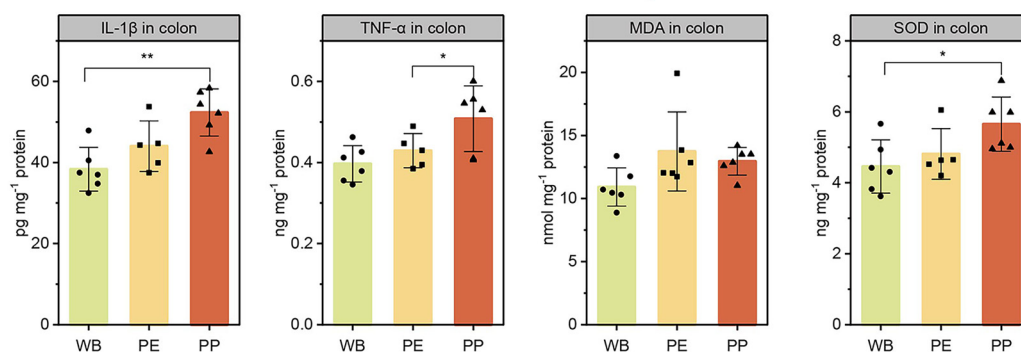


Figure 3. Effects of exposure to microplastics released from cutting boards on inflammatory factors and oxidative stress of mice (mean $\pm$ standard deviation, $n=6$). (A) Serum levels of C-reactive protein (CRP), pro-inflammatory cytokine tumor necrosis factor- α (TNF- α), anti-inflammatory cytokine interleukin-10 (IL-10), and carcinoembryonic antigen (CEA) in mice fed diets prepared on willow wooden (WB), polyethylene (PE), and polypropylene (PP) cutting boards for 4 wk. (B) Serum levels of CRP, TNF- α , IL-10, and CEA in mice fed diets prepared on WB, PE, and PP cutting boards for 12 wk. (C) Levels of pro-inflammatory cytokines IL-1 β and TNF- α , as well as malondialdehyde (MDA) and superoxide dismutase (SOD), in the ileum after exposure to diets prepared on WB, PE, and PP cutting boards for 12 wk. (D) Levels of IL-1 β , TNF- α , MDA, and SOD in the colon after exposure to diets prepared on WB, PE, and PP cutting boards for 12 wk. Statistical analyses were conducted using GraphPad Prism (version 9.0.0), employing one-way ANOVA with Tukey's test. * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$. The corresponding data are presented in Excel Table S2. Note: ANOVA, analysis of variance.

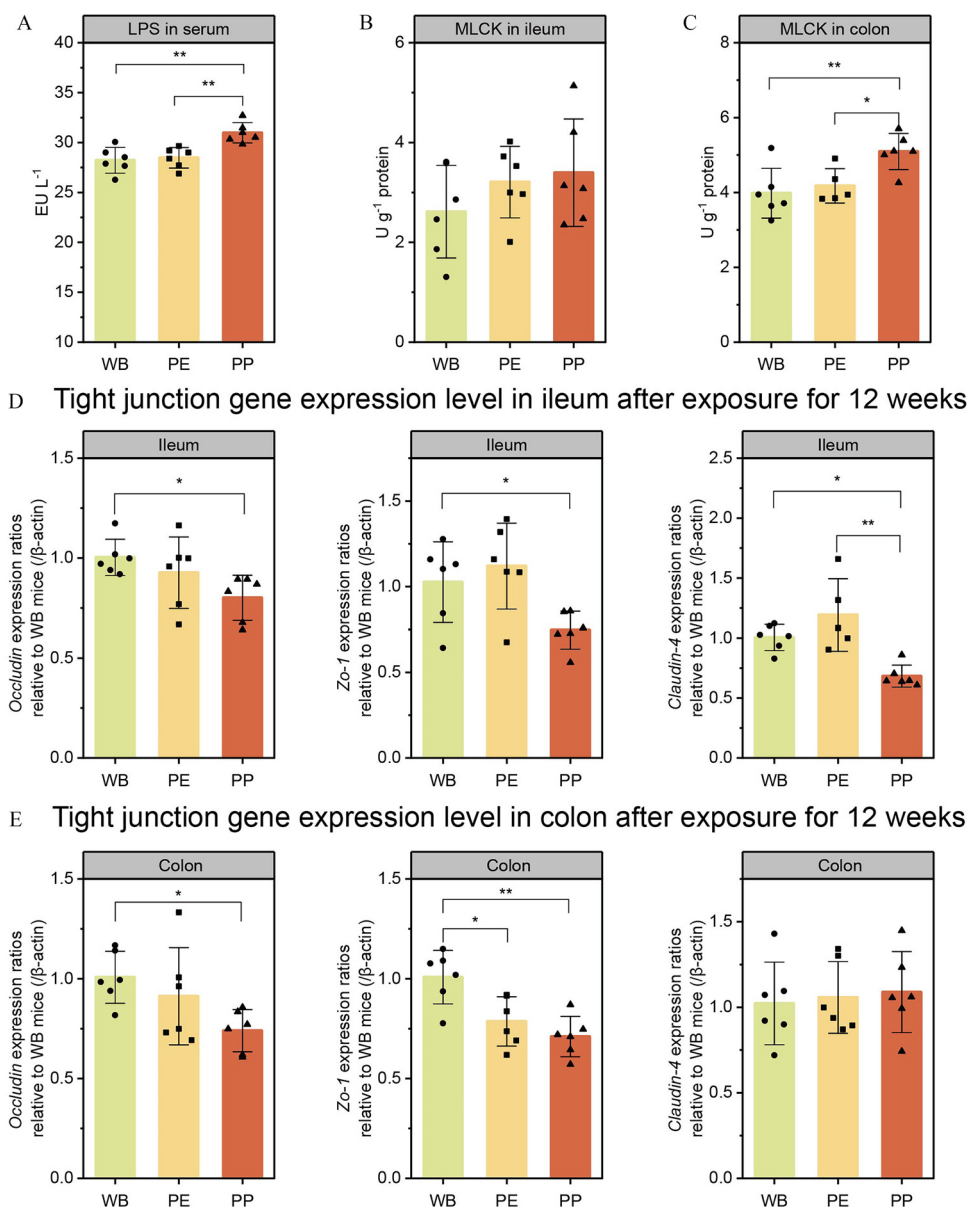


Figure 4. Effects of exposure to microplastics released from cutting boards on mouse intestinal tight junctions (mean $\pm$ standard deviation, $n=6$). (A) Serum level of lipopolysaccharide (LPS, an endotoxin) in mice fed diets prepared on willow wooden (WB), polyethylene (PE), and polypropylene (PP) cutting boards for 12 wk. Expression of myosin light chain kinase (MLCK) in the (B) ileum and (C) colon of WB, PE, and PP mice following 12 wk of exposure. Relative mRNA expression levels of *Occludin*, *Zo-1*, and *Claudin-4* that coded tight junction proteins Occludin, ZO-1, and Claudin-4 in the (D) ileum and (E) colon of WB, PE, and PP mice following 12 wk of exposure. Statistical analyses were conducted using GraphPad Prism (version 9.0.0), employing one-way ANOVA with Tukey's test. * $p < 0.05$ and ** $p < 0.01$. The corresponding data are presented in Excel Table S3. Note: ANOVA, analysis of variance.

0.0259), MDA (11.0 ± 2.07 vs. 7.36 ± 1.95 nmol/mg protein, $p = 0.0185$), and SOD (4.46 ± 0.94 vs. 3.06 ± 0.73 ng/mg protein, $p = 0.0293$) (Figure 3C); colon IL-1 β (52.3 ± 5.81 vs. 38.3 ± 5.40 pg/mg protein, $p = 0.0025$), TNF- α (0.51 ± 0.08 vs. 0.40 ± 0.04 ng/mg protein, $p = 0.0157$), and SOD (5.65 ± 0.77 vs. 4.46 ± 0.75 ng/mg protein, $p = 0.0376$) (Figure 3D); and serum LPS levels (31.0 ± 1.02 vs. 28.2 ± 1.29 EU/L, $p = 0.0019$), and colon MLCK activity (5.09 ± 0.48 vs. 3.98 ± 0.66 U/g protein, $p = 0.0090$) compared with those fed diets prepared on WB cutting boards (Figure 4A–C). In comparison, these mice exhibited 20.2%–32.0% ($p < 0.05$) lower relative expression of *Occludin*, *Zo-1*, and *Claudin-4* in the ileum and 26.6–29.5% lower relative expression of *Occludin* and *Zo-1* in the colon compared with mice fed diets prepared on WB cutting boards (Figure 4D,E).

In contrast, mice fed diets prepared on PE cutting boards did not show higher serum TNF- α and IL-10 levels compared with those fed diets prepared on WB cutting boards after 4 or 12 wk (Figure 3A,B). In addition, exposure to diets prepared on PE cutting boards for 12 wk did not result in significant differences in IL-1 β , TNF- α , MDA, and SOD in the ileum and colon tissues (Figure 3C,D), serum LPS (Figure 4A) and CEA levels (Figure 3B), ileum and colon MLCK activity (Figure 4B,C), or gene expression of intestinal tight junctions (Figure 4D,E).

Colon Mucous Layer

Mice fed diets prepared on PE and PP cutting boards for 12 wk exhibited a significantly higher coverage rate of colonic mucus

compared with those fed diets prepared on WB cutting boards ($22.3 \pm 1.19\%$ and $28.2 \pm 1.73\%$ vs. $16.3 \pm 0.85\%$, $p = 0.0035$ and <0.0001 , respectively) (Figure S5A). Furthermore, exposure to diets prepared on PP cutting boards for 12 wk led to 2.77- and 1.75-fold higher expression of the *Muc2* gene in the ileum and colon and a higher concentration of sIgA in the colon (47.1 ± 5.29 vs. 37.3 ± 5.16 $\mu\text{g}/\text{mg}$ protein, $p = 0.0429$) compared with those fed diets prepared on WB cutting boards (Figure S5B–E).

Fecal Microbiota

Based on PCA, the fecal microbial communities of mice fed diets prepared on WB, PP, and PE cutting boards over 4 wk showed no separation (Figure 5A). However, by 12 wk, the fecal microbiota of mice fed diets prepared on PE cutting boards, but not those fed diets prepared on PP cutting boards, differed significantly from those fed

diets prepared on WB cutting boards. After 12 wk of exposure, mice fed diets prepared on PE cutting boards exhibited a lower relative abundance of Firmicutes (mean $67.6 \pm 13.1\%$ vs. $84.2 \pm 8.83\%$, $p = 0.0152$) and a higher abundance of Desulfobacterota ($22.5 \pm 22.0\%$ vs. $5.94 \pm 4.99\%$, $p = 0.0260$) compared with those fed diets prepared on WB cutting boards (Figure 5B,C). At the genus level, the relative abundance of beneficial bacteria such as *Lactobacillus* ($20.2 \pm 13.6\%$ vs. $36.6 \pm 17.6\%$, $p = 0.1797$), *Faecalibaculum* ($5.75 \pm 3.00\%$ vs. $22.5 \pm 22.4\%$, $p = 0.0931$), and *Bifidobacterium* ($0.50 \pm 0.59\%$ vs. $5.22 \pm 6.93\%$, $p = 0.1320$) tended to be lower following exposure to diets prepared on PE cutting boards, although these differences were not statistically significant. Conversely, after 12 wk of exposure to diets prepared on PE cutting boards, the abundance of *Desulfovibrio* ($21.9 \pm 13.8\%$ vs. $5.48 \pm 5.02\%$, $p = 0.0260$) was significantly higher compared with those fed diets prepared on WB cutting boards (Figure 5D,E).

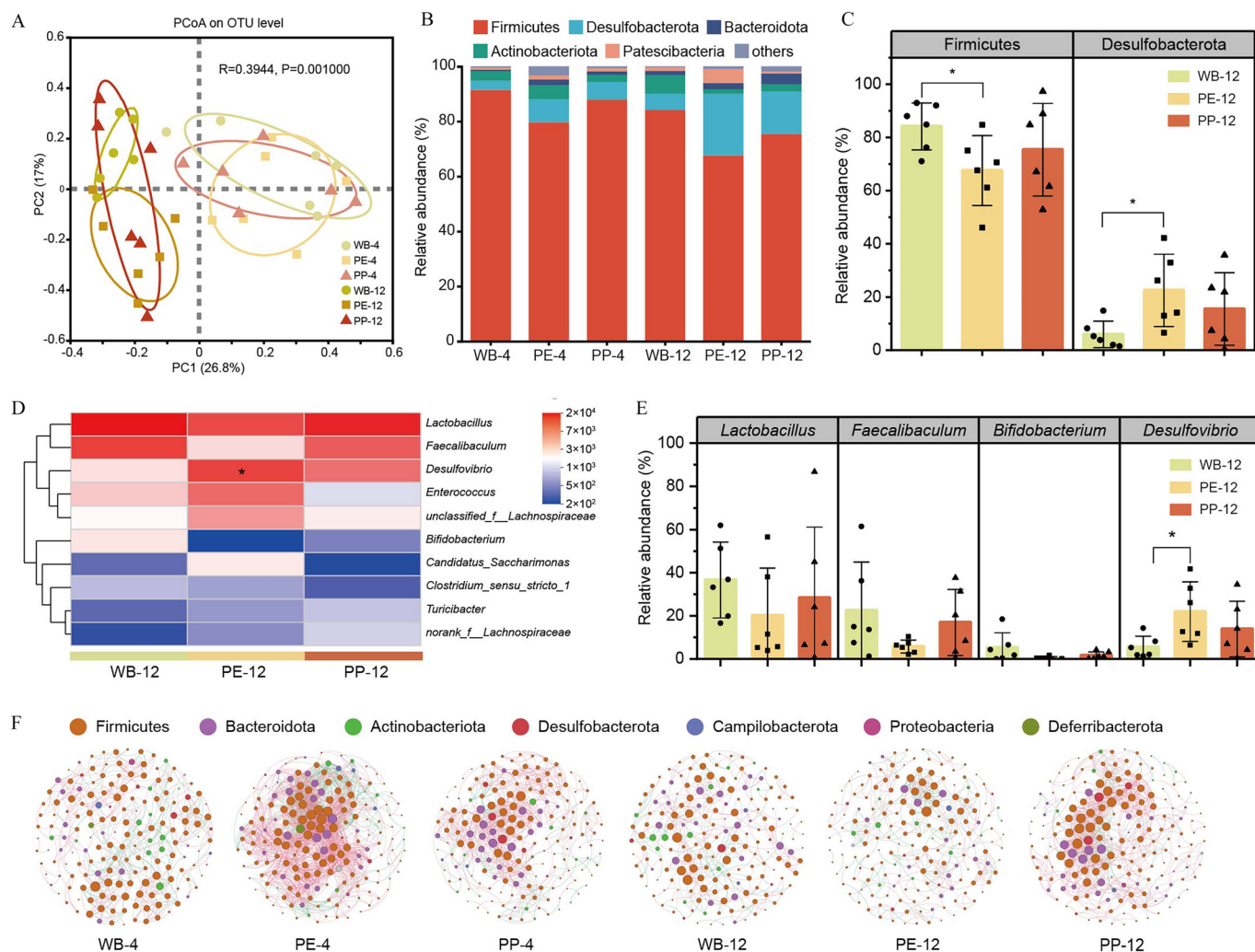


Figure 5. Effect of exposure to microplastics released from cutting boards on mouse fecal microbiota (mean $\pm$ standard deviation, $n = 6$). (A) UniFrac principal coordinates analysis (PCA) of the fecal microbial community of mice fed diets prepared on willow wooden (WB), polyethylene (PE), and polypropylene (PP) cutting boards for 4 wk (WB-4, PE-4, and PP-4) and 12 wk (WB-12, PE-12, and PP-12). (B) Bar plot showing fecal microbiota composition at the phylum level. (C) Relative abundances of Firmicutes and Desulfobacterota in the cecal contents of WB-12, PE-12, and PP-12 mice. (D) Heatmaps showing top 10 abundance of genera in the cecal contents of WB-12, PE-12, and PP-12 mice. (E) Relative abundances of *Lactobacillus*, *Faecalibaculum*, *Bifidobacterium*, and *Desulfovibrio* in the cecal contents of WB-12, PE-12, and PP-12 mice. (F) Network analysis showing associations among taxa for WB-4, PE-4, PP-4, WB-12, PE-12, and PP-12 mice. The network properties (e.g., number of nodes, number of edges, positive edge rate, negative edge rate, negative:positive cohesion, and modularity) indicating stability of gut microbial communities are shown in Table S4. In the networks, the nodes represent individual taxa and edges represent observed correlations in abundance among taxa from which interactions may be inferred, and the red and green edges represent positive and negative associations. Statistical analyses were conducted using GraphPad Prism (version 9.0.0), employing one-way ANOVA with Tukey's test. * $p < 0.05$. The corresponding data are presented in Excel Table S4. Note: ANOVA, analysis of variance; OTU, operational taxonomic unit; PC1, principal coordinate 1; PC2, principal coordinate 2.

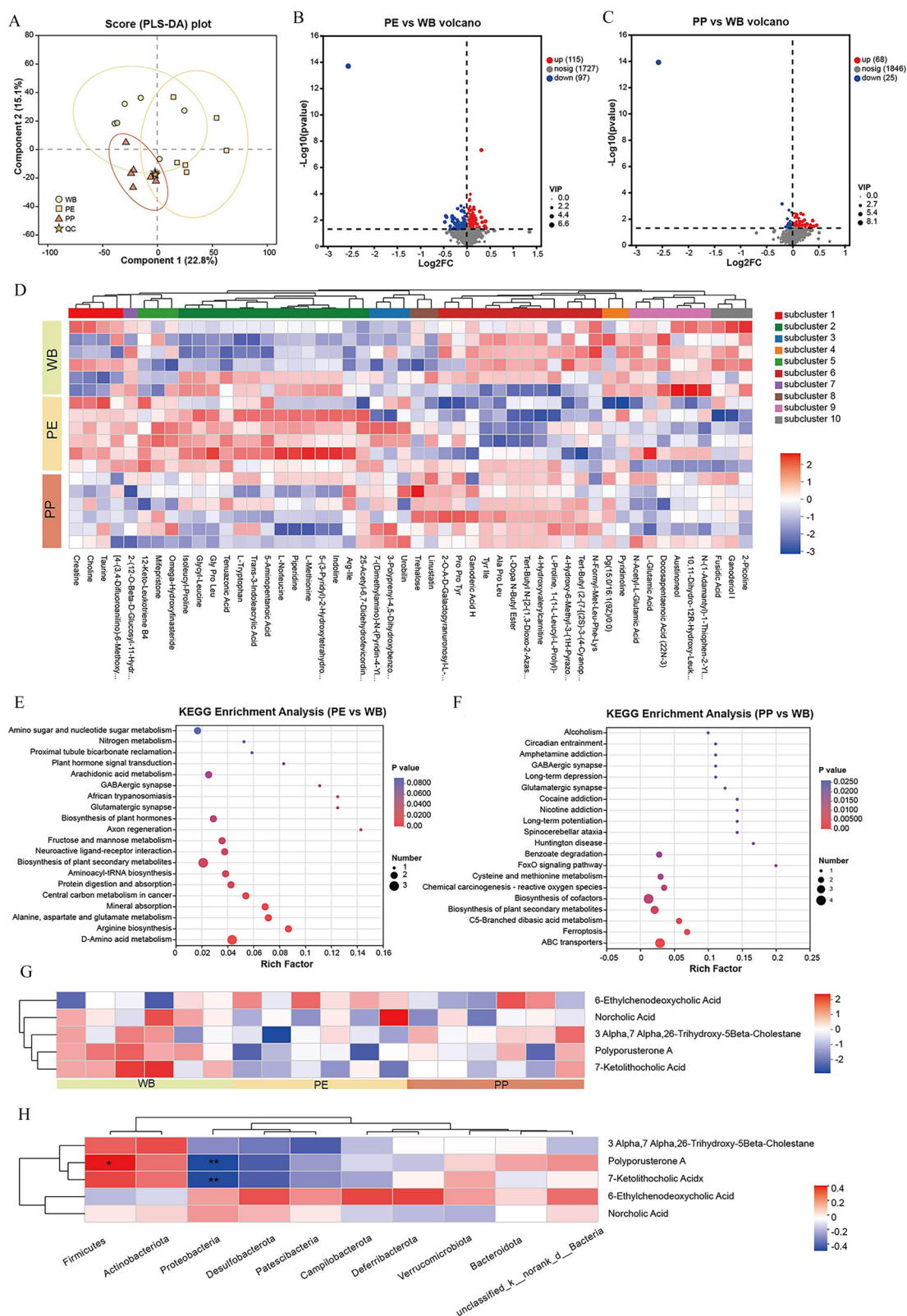


Figure 6. Effects of exposure to microplastics released from cutting boards on mouse fecal metabolomics. (A) Partial least squares discriminant analysis (PLS-DA) of metabolite composition in the cecal contents among mice fed diets prepared on willow wooden (WB), polyethylene (PE), and polypropylene (PP) cutting boards for 12 wk. Volcano plots showing the up-/down-regulated expression of metabolites in the cecal contents of (B) PE and (C) PP mice compared with WB mice. (D) Heatmap showing expression of top 50 abundance differential metabolites in the cecal contents of WB, PE, and PP mice. (E) KEGG enrichment analysis of differential fecal metabolites between PE and WB mice. (F) KEGG enrichment analysis of differential fecal metabolites between PP and WB mice. (G) Heatmap showing expression of bile acid differential metabolites including 6-ethylchenodeoxycholic acid, norcholic acid, 3 α , 7 α , 26-trihydroxy-5 β -cholestane, polyporusterone A, and 7-ketolithocholic acid in the cecal contents of WB, PE, and PP mice. (H) Spearman's correlation analysis of the relative abundance of bile acid differential metabolites and abundance of top 10 phyla in the cecal contents of WB, PE, and PP mice was performed in R (version 1.6.2; R Development Core Team) and is presented in a heatmap. * $p < 0.05$ and ** $p < 0.01$. The corresponding data are presented in Excel Table S5. Note: FC, fold change; KEGG, Kyoto Encyclopedia of Genes and Genomes; nosig, no significant change; QC, quality control; VIP, variable importance in projection.

During the initial 4 wk of exposure to diets prepared on PP and PE cutting boards, higher positive associations among taxa (80.1% and 86.3% vs. 64.7%), lower ratios of negative-to-positive associations (0.25 and 0.16 vs. 0.55), and lower modularity (0.43 and 0.59 vs. 0.76) were observed compared with the microbiota of mice fed diets prepared on WB cutting boards (Figure 5F; Table S4). After 12 wk of exposure to diets prepared on PP cutting boards, higher positive associations among taxa (87.4% vs. 70.9%) and lower modularity (0.57 vs. 0.79) remained compared with those fed diets prepared on WB cutting boards. Conversely, mice fed diets prepared on PE cutting boards showed recovery in modularity and ratios of negative-to-positive associations by the 12-wk mark.

Fecal Metabolomics

Based on PLS-DA, exposure to diets prepared on PP and PE cutting boards over 12 wk altered mouse fecal metabolomics (Figure 6A). Compared with mice fed diets prepared on WB cutting boards, 115 and 97 metabolites were respectively up- and down-regulated in mice fed diets prepared on PE cutting boards (Figure 6B). In contrast, 68 and 25 metabolites were respectively up- and down-regulated in mice fed diets prepared on PP cutting boards (Figure 6C). A heatmap showing the expression of the top 50 abundant differential metabolites exhibited more remarkable differences between mice fed diets prepared on PE and WB cutting boards than between mice fed diets prepared on PP and WB cutting boards (Figure 6D). Based on KEGG enrichment analysis, mice fed diets prepared on PE and PP cutting boards demonstrated distinct fecal metabolism pathway differences. D-Amino acid metabolism, arginine biosynthesis, and alanine, aspartate, and glutamate metabolism were the overall most significantly different pathways in the cecal contents of mice fed diets prepared on PE cutting boards. In contrast, ABC transporters, ferroptosis, and C5-branched dibasic acid metabolism were the most significantly different pathways in mice fed diets prepared on PP cutting boards (Figure 6E,F). Specifically, 4 bile acid metabolites (norcholeic acid, 3 α , 7 α , 26-trihydroxy-5 β -cholestane, polyporusterone A, and 7-ketolithocholic acid) were remarkably down-regulated in the cecal contents of mice fed diets prepared on PE cutting boards. In contrast, 6-ethylchenodeoxycholic acid was significantly up-regulated compared with those fed diets prepared on WB cutting boards (Figure 6G). Compared with PE MPs, the impacts of PP MPs on the expression of these bile acid metabolites were less evident. For the mice of the three groups, a significant positive correlation was observed between the bile acid metabolite polyporusterone A and the relative abundance of Firmicutes in the cecal contents, whereas the relative abundance of Proteobacteria was significantly and negatively related to polyporusterone A and 7-ketolithocholic acid (Figure 6H).

Liver Metabolomics

Based on PLS-DA of liver metabolomics, the mice fed diets prepared on PE cutting boards were significantly separated from the mice fed diets prepared on WB cutting boards, whereas the mice fed diets prepared on PP cutting boards showed some overlap with those fed diets prepared on WB cutting boards (Figure 7A). Compared with mice fed diets prepared on WB cutting boards, 117 and 90 metabolites were respectively up- and down-regulated in the liver of mice fed diets prepared on PE cutting boards (Figure 7B), whereas 93 and 68 metabolites were respectively up- and down-regulated in liver of mice fed diets prepared on PP cutting boards (Figure 7C). Figure 7D shows the expression of the top 50 abundant differential metabolites in the liver of mice fed diets prepared on WB, PE, and PP cutting boards.

Based on KEGG enrichment analysis, glycerophospholipid metabolism and sphingolipid signaling pathways were the overall most significantly different pathways by exposure to both PE and PP MPs (Figure 7E,F). Specifically, 6 differential metabolites associated with bile acid metabolism were identified, among which 3 α , 7 α , 12 α , 19-tetrahydroxy-5 β -cholanoic acid, sulfolithocholylcysteine, 24(28)dehydromakisterone, and deoxycholic acid were significantly down-regulated in mice fed diets prepared on PE cutting boards. In contrast, the down-regulation of these metabolites in mice fed diets prepared on PP cutting boards was less evident compared with the mice fed diets prepared on WB cutting boards (Figure 7G). For the mice of the three groups, significant positive correlations were observed between the down-regulated bile acid metabolites in the liver and the down-regulated bile acid metabolites in cecal contents (Figure 7H).

Discussion

Release of MPs during the Use of Plastic Cutting Boards

This study, based on the detection of MPs in mouse diets prepared using PE and PP cutting boards (Figure 2), suggests that the use of plastic cutting boards for food preparation can lead to the release of plastic particles, potentially exposing humans to MPs. Research estimates the annual per-person exposure at 7.4–50.7 g of MPs from PE cutting boards and 49.5 g from PP cutting boards.⁴³ Interestingly, the concentration of MPs in mouse diets increased with the number of cutting cycles (Figure 2G), indicating that the release of MPs from plastic cutting boards escalates with prolonged usage. Similar trends have been reported by Yadave et al.³⁷ An earlier study noted that a single cut on a new PP cutting board could release 100–300 MPs.³⁶ Data from this study showed that ~0.106 mg and 0.078 mg of MPs were released per cut from new PP and PE cutting boards, respectively, whereas 0.249 mg and 0.277 mg were released per cut during the 12th cutting cycle. With continued use, the release of MPs is expected to increase. These findings underscore that older plastic cutting boards may be more prone to releasing MPs during food preparation compared with newer ones.

This study highlights that both PE and PP cutting boards released significant quantities of MPs during use despite variations in the amount released and the size of the plastic particles. Although the total accumulated release of MPs from PP cutting boards was 26.5% lower than that from PE cutting boards (Figure 2H), the MPs from PP cutting boards were, on average, three times smaller in size (Figure 2C–F). Consequently, the number of MPs released from PP cutting boards was significantly higher than from PE cutting boards. Yadave et al.³⁷ showed that the mass and number of MPs released from PP cutting boards were greater than from PE cutting boards by 5%–60% and 14%–71%, respectively. The variance in the quantity and size of MPs released from different types of cutting boards could substantially affect their health implications.

Studies have detected MPs in the human body. For instance, a recent study detected MPs, including polyethylene terephthalate (PET) and PS plastics, in human blood, with particle sizes starting from 700 nm.⁸ Another study identified MP particles in the olfactory bulb of the human brain for the first time, with sizes ranging from 5.5 μ m to 26.4 μ m.⁴⁸ Although previous studies have reported the release of MPs from plastic cutting boards, none to our knowledge have evaluated the health impacts of such exposure through mouse bioassays. In this research, despite detecting very low concentrations of heavy metals and plastic additives in PE and PP cutting boards (Table S2, S3) and the absence of plastic particles in the tissues of the mouse jejunum, colon, and liver (Figure S4), mice fed diets prepared on these

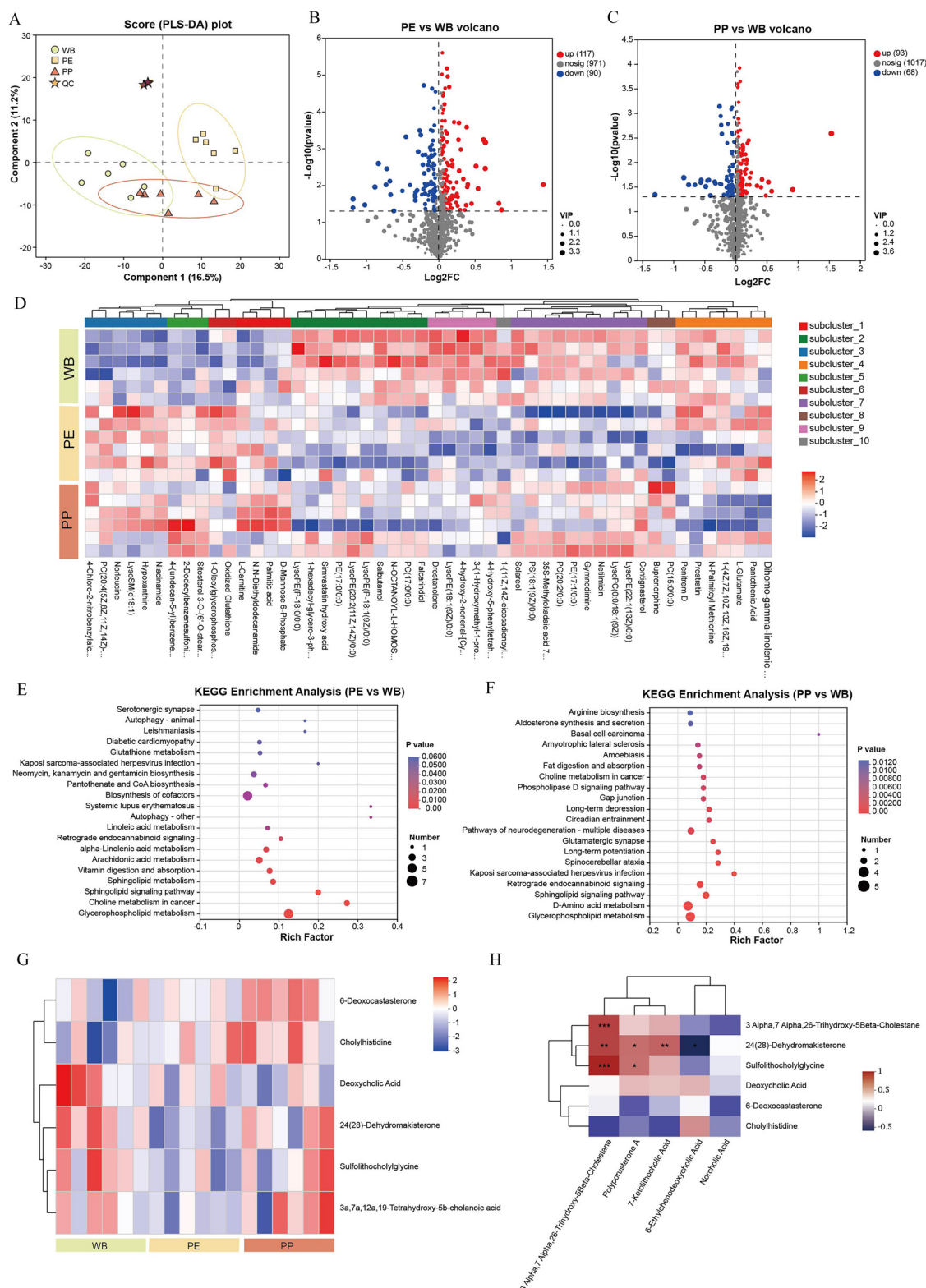


Figure 7. Effects of exposure to microplastics released from cutting boards on mouse liver metabolomics. (A) Partial least squares discriminant analysis (PLS-DA) of metabolite composition in the livers among mice fed diets prepared on willow wooden (WB), polyethylene (PE), and polypropylene (PP) cutting boards for 12 wk. Volcano plots showing the up/down-regulated expression of metabolites in the livers of (B) PE mice and (C) PP mice compared with WB mice. (D) Heatmap showing expression of top 50 abundance differential metabolites in the livers of in WB, PE, and PP mice. (E) KEGG enrichment analysis of differential gut metabolites between PE and WB mice. (F) KEGG enrichment analysis of differential gut metabolites between PP and WB mice. (G) Heatmap showing expression of bile acid differential metabolites including 6-ethylchenodeoxycholic acid, cholythistidine, deoxycholic acid, 24(28)-dehydromakisterone, sulfolithocholyglycine, and 3 α , 7 α , 12 α , 19-Tetrahydroxy-5 β -cholanoic acid in the livers of in WB, PE, and PP mice. (H) Spearman's correlation analysis of the relative abundance of six bile acid differential metabolites in the liver and the relative abundance of five bile acid differential metabolites in the cecal contents of WB, PE, and PP mice was performed in R (version 1.6.2; R Development Core Team) and is presented in a heatmap. * $p < 0.05$, ** $p < 0.01$ and *** $p < 0.001$. The corresponding data are presented in Excel Table S6. Note: FC, fold change; KEGG, Kyoto Encyclopedia of Genes and Genomes; nosig, no significant change; QC, quality control; VIP, variable importance in projection.

cutting boards still exhibited significant differences from mice exposed to diets prepared on wooden cutting boards with regard to serum inflammatory biomarkers, fecal microbiota, and fecal and liver metabolomics. Notably, diets prepared on different types of cutting boards were associated with distinct health outcomes: The mice fed diets prepared on PP cutting boards exhibited markers of intestinal inflammation, whereas those fed diets prepared on PE cutting boards exhibited differences in fecal microbiota, fecal and liver metabolomics. These findings suggest that neither PE nor PP cutting boards are safe in terms of release and health impacts of MPs.

Markers of Intestinal Inflammation

In this study, the mice exposed to PP MPs released from cutting boards exhibited significantly higher levels of inflammatory and cancer biomarkers in serum, as well as higher inflammatory biomarkers and oxidative stress in the ileum and colon compared with those fed diets prepared on wooden boards (Figure 3). Conversely, exposure to released PE MPs did not result in discernible differences in inflammation biomarkers. Under PS particle intake and interaction with the cellular microenvironment, localized inflammation in the intestine, as evidenced by elevated oxidative stress, has been reported.⁴⁹ The smaller particle size and higher number of MPs released from the PP cutting boards were likely contributors to the more pronounced intestinal inflammatory response. Previous studies have demonstrated that smaller MP particle sizes cause greater damage to organisms.⁵⁰

The composition of gut microbiota plays a pivotal role in host health, influencing various physiological processes, such as digestion, immune function, and disease resistance, and the gut microbiota acts as the primary intestinal barrier against external toxins.^{51,52} A balanced gut microbiota is characterized by a diverse array of microorganisms, predominantly bacteria, that engage in symbiotic interactions with the host. Factors including diet, genetics, age, and environmental exposures can significantly alter microbial diversity and balance, potentially leading to adverse health outcomes.⁵³ Research has demonstrated that alterations in this composition, often referred to as dysbiosis, are linked to numerous conditions such as inflammatory bowel disease, obesity, diabetes, and even mental health disorders.⁵² However, in this study, although the mice exposed to PP MPs did not demonstrate significant differences in the gut microbiota (Figure 5), higher levels of inflammation markers were observed in the ileum and colon tissues (Figure 3). This suggests that the intestinal inflammation associated with PP MP exposure may not be directly related to changes in gut microbiota.

In the intestine, the mucous layer covering the intestinal surface acts as another critical component of the intestinal barrier, effectively reducing the inflammatory response and playing a vital role in maintaining mucosal homeostasis.^{32,54} Damaged mucous layers are often linked to an increased risk of intestinal inflammation.⁵⁵ However, in this study, mice exposed to PP MPs demonstrated more intestinal mucus and higher expression of sIgA compared with those fed diets prepared on wooden boards, potentially due to a significantly higher expression of the *Muc2* gene in the ileum and colon (Figure S5), which strengthens the mucous layer. Djouina et al.⁵⁶ observed larger mucosal and mucous areas and an up-regulation of *Muc2* in mice after 6 wk of exposure to PE microbeads. The enhanced mucous layer may be a response to the heightened risk of intestinal inflammation, potentially triggering a feedback mechanism that stimulates the secretion of mucin and sIgA to mitigate MP-induced inflammation (Figure S5). sIgA is produced by plasma cells in the lamina propria of the intestinal mucosa and is a principal component of intestinal mucosal acquired immunity, functioning to inhibit bacterial

adhesion, prevent antigen absorption, and neutralize toxins.^{57,58} The enhanced mucous barriers contrast the higher expression of inflammatory markers in the ileum and colon following exposure to MPs from PP cutting boards (Figure 3), suggesting that the associated intestinal inflammation is not attributable to effects on the mucous barrier functions.

In comparison, the lower expression of specific tight junction mRNA (Figure 4), which can indicate intestinal tight junction damage, leading to enhanced intestinal permeability and allowing paracellular flux of LPS and other luminal antigens, may be associated with elevated inflammation biomarkers in the ileum and colon after exposure to PP MPs. After a 12-wk exposure to PP MPs, higher serum LPS levels (Figure 4A) suggest disturbed tight junctions and increased intestinal permeability. This is further supported by significantly lower expression of the *Occludin* and *Zo-1* genes in both the ileum and colon (Figure 4D,E). LPS, complex amphiphilic molecules released from the cell walls of Gram-negative bacteria through shedding or bacterial lysis,^{59,60} are predominantly found in the gut lumen, home to trillions of commensal bacteria. Under normal conditions, LPS does not penetrate the healthy intestinal epithelium⁶¹; however, in disorders characterized by intestinal permeability, a defective tight junction barrier permits paracellular flux of LPS and other antigens.^{60,62–65} Elevated levels of intestinal tissue and circulating LPS, which are significant in inflammatory bowel disease, mediate the inflammatory response.^{59,60,63,65} Typically, higher permeability of paracellular tight junctions results from the activation of MLCK and phosphorylation of myosin II regulatory light chain, which triggers contraction of the perijunctional actomyosin ring and alters the cytoskeletal structure.⁶⁶ In this study, a 12-wk exposure to PP MPs significantly activated MLCK in the colon (Figure 4C), leading to greater intestinal permeability.

Increased intestinal permeability allows LPS to enter the bloodstream and initiate inflammation. Upon entering the bloodstream, LPS first binds to LPS-binding protein, which facilitates its transport to the membrane surface of immune cells. There, LPS attaches to CD14, a membrane protein,⁶⁷ which then conveys LPS to the Toll-like receptor 4 (TLR4) and myeloid differentiation protein 2 (MD2) complex, aiding TLR4 in recognizing LPS.⁶⁸ Binding of LPS to TLR's extramembrane motif induces a conformational change in its intramembrane motif, which transmits the signal into immune cells.^{67–69} Within the immune cell, activation of signaling molecules, such as myeloid differentiation factor 88, IL-1R-associated protein kinase (IRAK) with IRAK2, and tumor necrosis factor receptor activating factor 6, results in phosphorylation of the inhibitory protein kappa B (IB) kinase complex, degradation of IB, and subsequent activation of the transcription factor nuclear factor kappa-light-chain-enhancer of activated B cells (NF-κB).^{49,70} NF-κB then translocates into the nucleus and binds to specific chromosomal regions, facilitating the expression of certain genes and the production of various cytokines, thereby inducing an inflammatory response.

The risk of intestinal inflammation associated with PP MPs was further evidenced by assessing gut metabolomics. Abnormal gut metabolism can precipitate inflammation and potentially lead to cancer.⁷¹ Notably, the abundance of L-glutamic acid, linked to cancer,⁷² was significantly higher in the guts of mice exposed to PP MPs compared to those exposed to diets prepared on WB cutting boards (Figure S6), potentially explaining the elevated CEA levels in serum, a biomarker of cancer risk (Figure 3B). Conversely, the levels of riboflavin—a water-soluble vitamin from the B family crucial for mitochondrial energy production and reducing inflammation and oxidative stress—were significantly lower in the guts of these mice (Figure S6).⁷³ This decrease in riboflavin suggests a diminished anti-inflammatory capacity, likely contributing to heightened inflammation.

Gut Microbiota, Gut Metabolomics, and Liver Metabolomics

In contrast to PP MPs, the mice fed diets prepared on PE cutting boards exhibited no evident differences in inflammatory markers compared with those fed diets prepared on the WB cutting boards (Figure 3). However, significant differences in gut microbiota and metabolomics of both gut and liver were observed after a 12-wk exposure to PE MPs, compared with PP MPs, based on PCA and PLS-DA (Figures 5–7). Gut microbiota dysbiosis has been associated with an increased risk of various host diseases, potentially linked to an immune–inflammatory response triggered by a reduced production of short-chain fatty acids, especially butyric acid, and the accumulation of protein metabolites, histamine, and LPS.^{74–76} However, in the present study, despite significant alterations in gut microbiota, including a lower abundance of Firmicutes and a higher abundance of Desulfobacterota (Figure 5C), the 12-wk exposure to PE MPs did not result in evident intestinal inflammation. This may be attributed to the significant roles these changes play in mitigating gut metabolites. Notably, compared with mice fed diets prepared on the WB cutting boards, mice exposed to PE MPs showed up-regulated expression of certain anti-inflammatory metabolites in the gut, such as glutamine and the bile acid metabolite 6-ethylchenodeoxycholic acid (Figure 6G; Figure S6). Previous studies have indicated that glutamine is crucial for regulating tight junctions and maintaining barrier function, which can significantly reduce inflammation.^{77,78} Meanwhile, 6-ethylchenodeoxycholic acid has been shown to activate the Farnesoid X receptor, thereby diminishing inflammation and oxidative stress.⁷⁹

The specific alterations in gut microbiota resulting from long-term exposure to PE MPs may be linked to their pronounced effects on liver metabolism, particularly bile acid metabolism via the gut–liver axis. Bile acids, crucial for gut barrier function and inflammation, are synthesized in the liver from cholesterol and transported to the intestine, aiding in the absorption of dietary lipids and fat-soluble vitamins.⁸⁰ In the present study, the levels of bile acid metabolites were significantly reduced in the livers of

mice exposed to PE MPs (Figure 7G), suggesting that diets prepared on PE cutting boards may lead to decreased bile acid production in the liver due to MP exposure. This reduction in bile acid metabolites and subsequent intestinal entry could significantly alter the biotransformation of bile acids in the intestinal lumen by gut microbes, potentially contributing to the observed changes in gut microbiota.⁸¹ This hypothesis is supported by the observed decrease in the relative abundance of bile acid metabolites in the guts of mice exposed to PE MPs (Figure 6G), along with significant positive correlations between the reduced bile acid metabolites in the liver and those in the gut (Figure 7H) and a positive relationship between the decreased bile acid metabolites in the gut and the reduced relative abundance of Firmicutes (Figure 6H).

The relationship between bile acids and the intestinal microbiota is intricate and closely intertwined, with both playing crucial roles in maintaining intestinal equilibrium and defending against pathogens.^{81,82} The microbiota modulates the bile acid pool, which promotes intestinal health and renewal, whereas bile acid-derived microbial metabolites help prevent the invasion of harmful gut bacteria.^{80,82} Dysregulation of the intestinal microbial community can lead to a decrease in secondary bile acids and a reduction in the relative abundance of Firmicutes, potentially increasing the risk of colon cancer. Compared with healthy people, patients with colorectal cancer have been found to have lower levels of Firmicutes.⁸³ In the present study, although mice exposed to PE MPs for 12 wk did not exhibit signs of episodic inflammation, the decreased abundance of Firmicutes and secondary bile acids in the gut represents a significant health concern, particularly for humans who use PE cutting boards over extended periods.

Health Implications

Overall, the present study reveals that significant quantities of MPs can be released from both PP and PE cutting boards during food preparation, potentially increasing human MPs exposure risks.

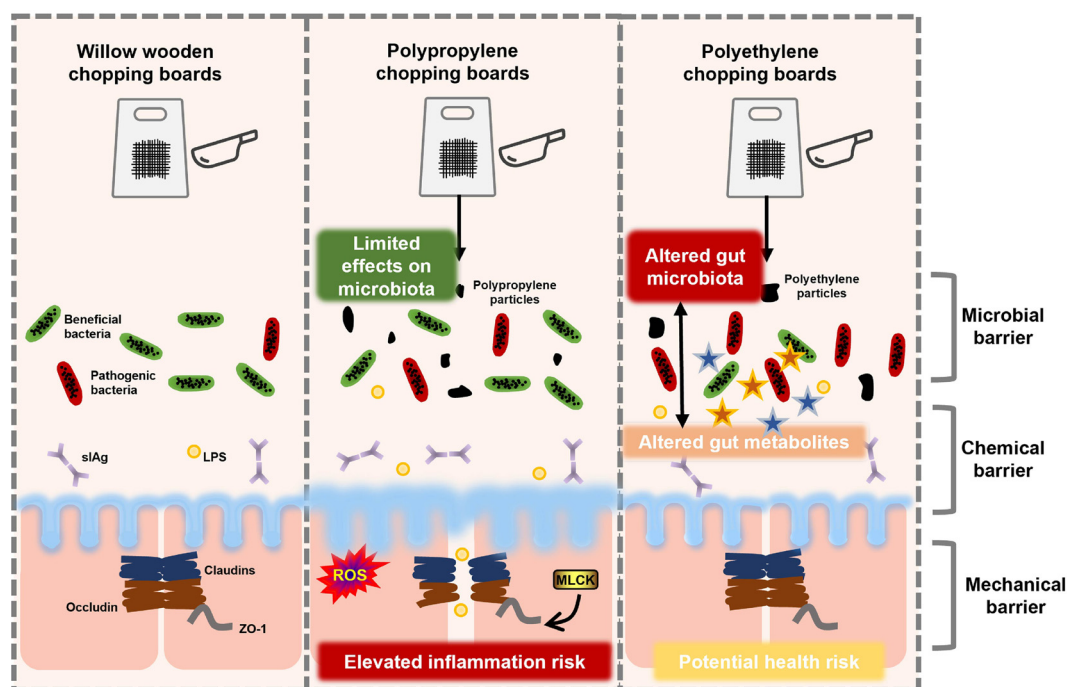


Figure 8. Proposed working model for the effect of long-term exposure to microplastics released from different plastic cutting boards on intestinal inflammation and gut microbiota. Note: LPS, lipopolysaccharide; MLCK, myosin light chain kinase; ROS, reactive oxygen species; sIgA, secretory immunoglobulin A.

Based on data in this study, we hypothesize a mechanistic understanding of the effects of long-term exposure to MPs released from different plastic cutting boards on intestinal inflammation and gut microbiota (Figure 8). Specifically, exposure to PP MPs, characterized by smaller particle sizes and higher numbers, may have limited effects on fecal microbiota but can damage paracellular tight junctions by activating MLCK. This damage increases intestinal permeability, allowing more endotoxins, such as LPS, to enter the bloodstream and potentially trigger intestinal inflammation (Figure 4). Conversely, exposure to PE MPs appears to have minimal effects on intestinal tight junctions and poses a low risk of intestinal inflammation. However, it can significantly alter fecal microbiota, fecal metabolism, and liver metabolism, likely through the gut–liver axis, representing another significant health risk (Figure 8). Consequently, this study highlights the varied health effects of MPs released from different cutting board materials, suggesting that no plastic cutting boards can be considered entirely “safe.” Considering the longevity of plastic cutting boards, which can span several years, we recommend minimizing their use to reduce human exposure to MPs. When their use is unavoidable, it is advisable to replace them regularly, given that older boards are more prone to releasing MPs during food preparation compared with newer ones.

The safety of plastic cutting boards for food preparation warrants further investigation. The interplay between the microbiota and the gut–liver axis metabolism is mutually influential. The observed changes in the fecal microbial community in this study may stem from a disrupted gut microbial balance due to abnormal metabolism within the gut–liver axis. In addition, the increased disruption in gut metabolism could be attributed to changes in the utilization and production of metabolites by microorganisms following shifts in the microbial balance, necessitating further research to clarify these relationships. Moreover, the health risks associated with MP exposure from plastic cutting boards were evaluated using a mouse model in this study. However, it is crucial to acknowledge the differences in microbiota between mice and human populations. The relevance of the health implications observed in mice to human populations requires further assessment. In this study, mice were exposed to relatively high concentrations of MPs released from plastic cutting boards for 12 wk. Future studies should explore the health effects of prolonged exposure to lower doses of MPs. Furthermore, although MPs were not detected in ileum, colon, and liver tissues of mice in this research, it remains to be determined whether MPs from plastic cutting boards can enter human tissues and organs. Given that the use of plastic cutting boards invariably leads to the release of MPs and causes multi-level health effects, developing dietary strategies to mitigate this toxicity is essential. Enhancing intestinal barrier functions by improving gut microbiota could be a promising strategy to reduce the health effects of MP exposure, which should be investigated in subsequent studies.

Acknowledgments

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